VAMSI KATTERA

Portfolio | **Gmail** | **GitHub** | **linkedin** | **07899877443**

Professional Summary

AI Engineer (MSc Advanced Computer Science, University of Strathclyde) with a strong Python foundation, who builds LLM and RAG systems and measures them against expert-labelled data. Built Obelus, a contract-review API that quotes the exact clause text and says "not found" when it cannot, evaluated on 510 real contracts (span F1 up from 22–46% to 59–75% on held-out tests), and a GOV.UK RAG assistant (95% of answers judged correct, 10/10 out-of-scope refused). Both are tested, Dockerised FastAPI services with CI and live demos. Also built deep learning and computer vision applications. Keen to turn business problems into measurable AI tools.

Education

MSc Advanced Computer Science – University of Strathclyde **Jan 2025 – Jan 2026**
Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence
BSc Computer Science & Engineering – KHIT, Andhra Pradesh, India **2020 – 2024**
CGPA: 7.66 | Relevant Coursework: Python, Data Structures & Algorithms, AI, ML

Skills

Languages: Python, SQL
LLM & RAG: RAG pipelines, hybrid retrieval (BM25 + dense), cross-encoder reranking, embeddings, prompt engineering, grounded answers with citations, LLM evaluation (LLM-as-judge), Gemini API, OpenAI API, agent workflows and tool integration, structured output (Pydantic), quote verification, evaluation against expert labels
Vector Search: Chroma, sentence-transformers
Machine & Deep Learning: TensorFlow, Keras, scikit-learn, CNNs, transfer learning, RNNs, LSTMs, Transformers, model evaluation, hyperparameter tuning, feature engineering
Computer Vision: YOLOv8, OpenCV, object detection, object tracking, video analytics
Backend & APIs: FastAPI, REST API design, request validation, rate limiting, structured logging, metrics
MLOps & Deployment: Docker, GitHub Actions CI, pytest, Hugging Face Spaces, AWS EC2, Render
Tools: Git, GitHub, Streamlit, NumPy, Pandas
Foundations: Linear Algebra, Calculus, Probability & Statistics

AI Projects

Obelus – Evaluated Contract-Review API (Cite or Abstain) | Live Demo | GitHub
Tech Stack: Python, FastAPI, Gemini API, BM25, Pydantic, Docker, GitHub Actions, Hugging Face Spaces

- Built an API that finds clauses in contracts and returns the exact passages with character positions, checking that every quote exists word for word in the contract and answering "not found" when it does not.
- Measured it on 510 expert-labelled contracts (CUAD), split by contract: on 102 held-out test contracts, LLM quoting raised span F1 over a BM25 baseline from 22–46% to 59–75% across four clause types.
- Diagnosed 31% recall on liability caps by sorting every miss (search, model refusal, quote not found); the cause was a clause definition narrower than the experts', and fixing it, not the model, raised the score.

- Reported what did not work: presence F1 on liability caps fell from 91% to 86%, and a BM25 gating step gave no gain, so I kept the simpler method.
- Shipped it as a tested FastAPI service (55 tests, CI, non-root Docker) with a web demo on Hugging Face Spaces, input limits and per-visitor and daily rate limits.

GOV.UK RAG Assistant – Cited Q&A on Government Guidance | Live Demo | GitHub
Tech Stack: Python, FastAPI, Gemini API, Chroma, BM25, cross-encoder reranking, Streamlit, Docker, GitHub Actions, Hugging Face Spaces

- Built a RAG assistant over 129 GOV.UK pages (visas, right to work, benefits) that answers with citations to the source page and refuses when the pages do not cover the question.
- Combined vector and BM25 search with rank fusion and cross-encoder reranking, raising top-1 retrieval accuracy from 74% to 86% on a hand-reviewed 42-question test set.
- Added a similarity gate and a cite-or-abstain rule: 10/10 out-of-scope questions refused and 95% of in-scope answers judged correct (LLM judge, spot-checked by hand).
- Shipped it as a tested FastAPI service (pytest, CI, rate limiting, JSON logs, metrics) in Docker with a Streamlit UI, deployed on Hugging Face Spaces.

AI Traffic Monitoring System – Real-Time Computer Vision
Tech Stack: Python, YOLOv8, ByteTrack, OpenCV, scikit-learn, FastAPI, React, Docker, Render

- Built a system that detects, tracks and counts vehicles from video or webcam feeds with YOLOv8 and ByteTrack, with lane-zone counts and a traffic density estimate.
- Trained an Isolation Forest to flag unusual traffic patterns and a Random Forest to classify traffic density.
- Counted vehicles crossing a line and per lane zone, and showed live counts, density and anomaly flags on a React dashboard.
- Served the video stream and analytics through a FastAPI service with a React dashboard, containerised with Docker and deployed on Render.

AI vs Real Image Classifier – Deep Learning Computer Vision
Tech Stack: Python, TensorFlow, Keras, MobileNetV2, Streamlit, AWS EC2

- Built a CNN classifier using transfer learning (MobileNetV2) to distinguish AI-generated images from real ones, covering preprocessing, training, validation and evaluation.
- Improved generalisation with data augmentation and dropout regularisation, reaching over 80% classification accuracy.
- Deployed as a real-time Streamlit application on AWS EC2 for live inference.

Certifications

Deep Learning Specialization – DeepLearning.AI (Coursera)
Mathematics for Machine Learning and Data Science – DeepLearning.AI (Coursera)